

CIS 121 Sequence Problems

Develop an IPO Chart and C++ code the following problems. Upload the IPO and code files to Blackboard.

Save your files with the convention PS2P1, PS2P2 etc. PS1P1 is Problem set 1, program 1 etc.

1. Allow the user to enter the quantity and unit price (price per item). Compute extended price (quantity x price). Display the extended price.

Input	Process	Output
• The quantity • The unit price	• Assign a variable to the input for the quantity • Assign a variable to the input of the unit price • Calculate the product of variables of quantity and unit price • Assign the product as extended price variable • Display the extended price	• The extended price

2. Allow the user to enter last name, hours and pay rate. Compute gross pay to be hours x rate. (Note: we are not giving time and a half for over time hours yet!). Display last name and gross pay.

Input	Process	Output
• User's last name • Hours • Pay rate	• Assign variable to user's last name • Assign a variable to the input Hours • Assign a variable to the user's (hourly) pay rate • Calculate the product of hours and pay rate • Assign the product as the variable gross pay • Display a message that includes the user's last name and gross pay	• User's last name • Gross pay

3. The user is to enter the length and width of a rectangle. Computer the area (length x width) and the circumference ($2 \times \text{length} + 2 \times \text{width}$). Display the area ad circumference.

Input	Process	Output
- Length - Width	- Assign the length as a variable - Assign the width as a variable - Calculate the product of the length and width - Assign the product as the variable "area" - Calculate $2 * \text{length} + 2 * \text{width}$ - Assign the result as the variable "circumference" - Display area and circumference	- Area - Circumference

4. Enter last name and credits taken. Tuition is \$250 per credit hour. Add a \$100 lab fee. Compute total tuition (credits taken x 250 + lab fee). Display last name and tuition.

Input	Process	Output
- User's last name - Credits taken	- Assign last name as a variable - Assign credits taken as a variable - Assign a variable for the Tuition/credit hour. Set it at 250 - Assign a variable for the Lab fee. Set it at 100 - Calculate the product of credits taken and Tuition/credit hour, then add the lab fee - Assign the result as the variable Total Tuition - Display last name and Total Tuition	- User's last name - Total Tuition

5. The price of an item and discount percent is entered into the program. Display the discount amount and discounted price of the item. Note: enter the discount percent in decimal form.

Input	Process	Output
• Price of item • Discount percentage	• Assign price of item as a variable • Assign the discount percentage as a variable • Convert the discount percentage into decimal by multiplying it to 0.01 • Assign the product as Decimal Discount Percentage • Find the product of price of item and DPP • Subtract the calculated product from price of item • Assign the difference as the Discounted item price variable	• Discount percentage • Discounted item price